

Class Field Theory

1 0908

Goal Describe all abelian extension of number fields.
i.e. inverse Galois problem for abelian groups.

Recall (Kronecker-Weber Theorem) Every abelian extension of the rational numbers is contained in a cyclotomic field.

What happens if we replace $\mathbb{Q}$ by a number field K ?

Denote $\text{Cl}(K)$ as the ideal class group of K . (Fractional ideals modulo principal ideals.)

Theorem 1.1: Let L be the maximal unramified abelian extension of a number field K . Then L/K is finite and $\text{Gal}(L/K) \cong \text{Cl}(K)$.
 L is called the Hilbert class field of K .

Recall Let $N > 1$ be an integer. ζ_N denotes a primitive N -th root of unity. Then

1. $\text{Gal}(\mathbb{Q}(\zeta_N)/\mathbb{Q}) \cong (\mathbb{Z}/N\mathbb{Z})^\times$.
2. Suppose $N \not\equiv 2 \pmod{4}$. Then p is ramified in $\mathbb{Q}(\zeta_N)$ if and only if $p \mid N$.
3. If p splits completely in $\mathbb{Q}(\zeta_N)$, then p is unramified, so the decomposition group is isomorphic to $\text{Gal}(\kappa(P)/\kappa(p))$.

The Frobenius element $\text{Frob}(P/p)$ is the unique element in the decomposition group that acts as $x \mapsto x^p$ on the residue field.

It is too restrictive to ask for unramified extensions.

So we ask for abelian extensions.

1.1 Review $\mathbb{A}_K, \mathbb{I}_K = \mathbb{A}_K^\times, \text{Cl}(K)$

Let I be a fractional ideal in K . Then I can be written as $I = P_1^{a_1} \dots P_r^{a_r}$, where P_i are prime ideals and $a_i \in \mathbb{Z}$.

So the group of fractional ideals is a free abelian group generated by the prime ideals.

$$\bigoplus_P \mathbb{Z}$$

and Cl is the quotient of $\bigoplus_P \mathbb{Z}$ by the image of the map $K^\times \rightarrow \bigoplus_P \mathbb{Z}$ given by $x \mapsto (v_P(x))_P$, where v_P is the valuation at P .

Let $U = \prod_{v:\text{inf}} K_v^\times \times \prod_{v:\text{fin}} \mathcal{O}_v^\times$. It is an open subgroup of $\mathbb{A}_K^\times$.

Then $\mathbb{A}_K^\times/U$ is exactly $\bigoplus_{v:\text{fin}} \mathbb{Z}$. So we have a surjective map $\mathbb{A}_K^\times \rightarrow \text{Cl}(K)$, and the kernel is $UK^\times$.

As the class group corresponds to the maximal unramified abelian extension, we want to shrink U to a smaller open subgroup to detect more abelian extensions.

[Ray class group]@Obsidian

Remark: This isomorphism shows that $\text{Cl}(K, \mathfrak{a})$ is finite:

By the isomorphism, it is discrete.

Also, it is a quotient of $\mathbb{A}_K^1/K^\times$, which is compact, so it is finite.

Example 1.2: $\text{Cl}(\mathbb{Q}, N) \cong (\mathbb{Z}/N\mathbb{Z})^\times$.

[計算 $\text{Cl}(\mathbb{Q}, N)$]@Obsidian

By Kronecker-Weber, we know that $\text{Gal}(\mathbb{Q}^{\text{ab}}/\mathbb{Q})$ is the inverse limit of $(\mathbb{Z}/N\mathbb{Z})^\times$ over N .

By above, $(\mathbb{Z}/N\mathbb{Z})^\times$ is a quotient of $C_{\mathbb{Q}} = \mathbb{A}_{\mathbb{Q}}^\times/\mathbb{Q}^\times$.

Class field theory is saying that we can construct a map from $C_K = \mathbb{A}_K^\times/K^\times$ to $\text{Gal}(K^{\text{ab}}/K)$ for any number field K , and the image of this map is dense.

Theorem 1.3 (Local class field theory): Let K be a local field. There is a unique continuous homomorphism

$$\rho_K : K^\times \rightarrow \text{Gal}(K^{\text{ab}}/K)$$

satisfying the following properties:

1. For any L/K finite abelian extension, the kernel of ρ_K is exactly $N_{L/K}(L^\times)$, the norm of $L^\times$ in $K^\times$.

$$\begin{array}{ccccc} K^\times & \longrightarrow & \text{Gal}(K^{\text{ab}}/K) & \twoheadrightarrow & \text{Gal}(K^{\text{un}}/K) \\ \downarrow & & \downarrow & \swarrow & \\ \mathbb{Z} & \longrightarrow & \text{Gal}(\overline{\mathbb{F}_q}/\mathbb{F}_q) & & \end{array}$$

- 2.
3. There is a 1-1 correspondence between U : an open subgroup of $\text{Gal}(K^{\text{ab}}/K)$ and V : open subgroup of finite index of $K^\times$ given by $U \mapsto \rho_K^{-1}(U)$.

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Definition (CSA): A (finite-dimensional) central simple algebra (CSA) over a field K is a finite-dimensional K -algebra R such that:

- R is a finite-dimensional K -algebra.
- R is central: The center of R is exactly K .
- R is simple: R has no nontrivial two-sided ideals.

Example 2.1: The matrix algebra $M_{n(K)}$ is a CSA over K .

Fact 2.2 (Wedderburn): Let A be a CSA over K . Then there's a central division algebra D s.t.

$$A \cong M_n(D)$$

for some $n \in \mathbb{Z}_{>0}$ and D is unique up to isomorphism.

Definition Let A, B be two CSAs over K . We say A and B are Brauer equivalent if there exist $m, n \in \mathbb{Z}_{>0}$ s.t.

$$A \otimes_K M_m(K) \cong B \otimes_K M_n(K)$$

i.e. their corresponding division algebras are isomorphic. The Brauer equivalence class of A is denoted by $[A]$.

The Brauer group $\text{Br}(K)$ is the set of Brauer equivalence classes of CSAs over K with the group operation given by tensor product over K .

Then $[A]^{-1} = [A^{\text{op}}]$.

The importance of the Brauer group in CFT:

If K_v is non-archimedean, then $\text{Br}(K_v) \cong \mathbb{Q}/\mathbb{Z}$ and it is canonical.

We can write $\text{Br}(K_v)$ explicitly as its element is the form (χ, a) where χ is a character of $\text{Gal}(K_v^{\text{ab}}/K_v)$ and $a \in K_v^\times$.

It help us define a map

$$\begin{aligned} \rho : K_v^\times &\rightarrow \text{Hom}(\text{Hom}(\text{Gal}(K_v^{\text{ab}}/K_v), \mathbb{Q}/\mathbb{Z}), \mathbb{Q}/\mathbb{Z}) \\ a &\mapsto (\chi \mapsto \text{inv}(a, \chi)) \end{aligned}$$

and the right hand side is $\text{Gal}(K_v^{\text{ab}}/K_v)$ given by Pontryagin duality.

3 Lecture 1: Group cohomology

Let G be a finite group.

Definition

1. M is a G -module if M is an abelian group with a G -action that is compatible with the group structure.

We also consider M as a $\mathbb{Z}[G]$ -module.

2. $f : M \rightarrow N$ is a G -module homomorphism if f is a map preserving the G -action.

Definition $C^n(\mathbb{Z}[G], A) = \text{Hom}(\mathbb{Z}[G]^{n+1}, A)$.

$\partial^n : C^n(\mathbb{Z}[G], A) \rightarrow C^{n+1}(\mathbb{Z}[G], A)$ is defined by

$$(\partial^n f)(g_0, \dots, g_{n+1}) = \sum_{i=0}^{n+1} (-1)^i f(g_0, \dots, \hat{g}_i, \dots, g_{n+1})$$

It defines a cochain complex $(C^n(\mathbb{Z}[G], A), \partial^n)$, and the cohomology of this complex is denoted by $H^n(G, A)$.

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Definition A G -module B is called relatively injective if B is a direct summand of an induced module $\text{Ind}_1^G(A)$ for some abelian group A .

Corollary 4.1: A relatively injective G -module is acyclic.

4.1 Change of groups

General: $f : G' \rightarrow G$: group homomorphism of finite groups.

Let A be a G -module. Through f , we can make A a G' -module. Denote as f^*A . This is a functor $f^* : G\text{-Mod} \rightarrow G'\text{-Mod}$. We have a natural map $f^* : H^n(G, A) \rightarrow H^n(G', f^*A)$ for all n .

Now consider the case $H \hookrightarrow G$. The pullback map is called the restriction map.

Theorem 4.2 (Shapiro lemma):

$$H^n(H, A) \cong H^n(G, \text{Ind}_H^G(A))$$

Proof: By Frobenius reciprocity, we have

$$\text{Hom}_G(B, \text{Ind}_H^G(A)) \cong \text{Hom}_H(B, A)$$

For $n = 0$: Take $B = \mathbb{Z}$ with trivial action.

Now for $n > 0$, take a injective resolution $A \rightarrow I^\bullet$. By Frobenius reciprocity and the exactness of induction functor, $\text{Ind}_H^G(I^\bullet)$ is an injective resolution of $\text{Ind}_H^G(A)$.

Applying the case $n = 0$ to $I^\bullet$, we get the desired isomorphism since both sides are universal δ -functors.

■

Consider another case $H \hookrightarrow G$ and A is a G -module. Consider the following map:

For $n = 0$, we have a map $H^0(H, A|_H) \rightarrow H^0(G, A)$ given by $a \mapsto \sum_{g \in G/H} ga$.

Now, $H^n(H, A|_H)$ is a δ -functor. We want to show that it's the derived functor of $A \mapsto A^H = H^0(H, A|_H)$.

By the dual of Frobenius reciprocity,

$$\mathrm{Hom}_H(- \otimes_H G, I) \cong \mathrm{Hom}_G(-, I)$$

Since $- \otimes_H G$ is exact, $I^\bullet|_H$ is an injective resolution of $A|_H$. (Restriction is exact.)

This proves that $H^n(H, A|_H)$ is the derived functor of $A \mapsto A^H$ hence a universal δ -functor.

As $A \mapsto H^n(G, A)$ is also a δ -functor, we have a natural transformation $H^n(H, A|_H) \rightarrow H^n(G, A)$ for all n .

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5.1 Inflation

Let $H \triangleleft G$ be a normal subgroup.

Given a G -module A , we can make A^H a G/H -module.

Then we have a natural map

$$\mathrm{inf}_{G/H} : H^n(G/H, A^H) \rightarrow H^n(G, A^H) \rightarrow H^n(G, A)$$

Theorem 5.1: We have an exact sequence

$$0 \longrightarrow H^1(G/H, A^H) \xrightarrow{\mathrm{inf}_{G/H}} H^1(G, A^H) \xrightarrow{\mathrm{res}} H^1(H, A)$$

5.2 Intermezzo: Spectral sequences (arising from a double complex)

Set-up: $M^{\bullet, \bullet}$: bounded below double complex of abelian groups.

$$\begin{array}{ccccccc} & & 0 & & 0 & & 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & M^{0,0} & \longrightarrow & M^{0,1} & \longrightarrow & M^{0,2} \longrightarrow \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & M^{1,0} & \longrightarrow & M^{1,1} & \longrightarrow & M^{1,2} \longrightarrow \\ & & \downarrow & & \downarrow & & \downarrow \end{array}$$

with

- $d_{\rightarrow} \circ d_{\rightarrow} = 0$
- $d_{\downarrow} \circ d_{\downarrow} = 0$
- $d_{\rightarrow} \circ d_{\downarrow} + d_{\downarrow} \circ d_{\rightarrow} = 0$

Define the total complex $M^\bullet$ by $M^n = \sum_{p+q=n} M^{p,q}$ and $d = d_{\rightarrow} + d_{\downarrow}$.

We want to compute the cohomology of $M^\bullet$.

The strategy is to filter $M^\bullet$ by p and q and get two spectral sequences.

Define $F^i M^n = \sum_{p \geq i} M^{p, n-p}$.

Then

$$F^i H^n = (Z^n + F^i M^n) + B^n / B^n \subset H^n$$

It gives a filtration of H^n :

$$0 \subset F^n H^n \subset F^{n-1} H^n \subset \dots \subset F^0 H^n = H^n$$

- Page 0: $E_0^{p,q} = M^{p,q}$
- Page 1: $E_1^{p,q} = H^q(M^{p,\bullet})$ with the differential $d_1^{p,q} : E_1^{p,q} \rightarrow E_1^{p+1,q}$ induced by $d_{\rightarrow}$.
- Page 2: $E_2^{p,q} = H^p(E_1^{\bullet,q})$ with the differential $d_2^{p,q} : E_2^{p,q} \rightarrow E_2^{p+2,q-1}$ induced by $d_{\downarrow}$.

eye The map is given by:

For $[x] \in E_2^{p,q}$, we have $x \in M^{p,q}$ s.t. $d_{\rightarrow} x = 0$ and $d_{\downarrow} x = d_{\rightarrow}(-y)$, for some $y \in M^{p+1,q-1}$. Then $d_2^{p,q}([x]) = d_1^{p+1,q-1}([y])$.

Inductively, we can define

$$E_r^{p,q} = \ker(d_r^{p,q}) / \text{im}(d_r^{p-r,q+r-1})$$

and

$$d_r^{p,q} : E_r^{p,q} \rightarrow E_r^{p+r,q-r+1}$$

For a fixed (p, q) , the sequence $E_r^{p,q}$ stabilizes for r large enough. Denote the stable value by $E_{\infty}^{p,q}$.

If for some $x \in M^{p,q}$, we have $d_{\rightarrow} x = 0$ and $d_{\downarrow} x = 0$, i.e. $dx = 0$. Then $d_n(x) = 0$ for all n , so $[x] \in E_{\infty}^{p,q}$.

Similarly, if $x \in M^{p,q}$, $y \in M^{p+1,q-1}$ and $d(x+y) = 0$, then $[x+y] \in E_{\infty}^{p,q}$.

In fact,

$$E_{\infty}^{p,q} = \frac{Z^n \cap F^p M^{p+q}}{Z^n \cap F^{p+1} M^{p+q} + B^n \cap F^p M^{p+q}}$$

which is exactly the p -th graded piece of the filtration of H^n i.e. $\frac{F^p H^n}{F^{p+1} H^n}$.

Now, for lower degree terms, we have

- $F^1 H^1 = E_{\infty}^{1,0}$
- $H^1 / F^1 H^1 = E_{\infty}^{0,1}$
- $F^2 H^2 = E_{\infty}^{2,0}$

i.e.

$$0 \longrightarrow E_{\infty}^{1,0} \longrightarrow H^1 \longrightarrow E_{\infty}^{0,1} \longrightarrow 0$$

We have $E_{\infty}^{1,0} = E_2^{1,0}$ and $E_{\infty}^{0,1} = \ker(d_2^{0,1})$.

Also, $E_{\infty}^{2,0} = \text{coker}(d_2^{0,1})$.

So we have an exact sequence

$$0 \longrightarrow E_2^{1,0} \longrightarrow H^1 \longrightarrow E_2^{0,1} \xrightarrow{d_2} E_2^{2,0} \longrightarrow H^2$$

Theorem 5.2 (Grothendieck spectral sequence): Let $\mathcal{A}, \mathcal{B}, \mathcal{C}$ be three abelian categories.

$F : \mathcal{A} \rightarrow \mathcal{B}$ and $G : \mathcal{B} \rightarrow \mathcal{C}$ be two additive left-exact functors.

Assume that

- $\mathcal{A}, \mathcal{B}$ have enough injectives.
- F sends injective objects of $\mathcal{A}$ to G -acyclic objects of $\mathcal{B}$.

Then

$$E_2^{p,q} = (R^p G)(R^q F)(A) \implies R^{p+q}(G \circ F)(A)$$

For $A \in \mathcal{A}$, take an injective resolution $A \rightarrow I^\bullet$.

Then $F(I^\bullet)$ is a acyclic complex in $\mathcal{B}$.

Fact 5.3: There is a double complex $J^{p,q}$ s.t. $J^{p,\bullet}$ is an injective resolution of $F(I^p)$ for all p .

Example 5.4:

- $\mathcal{A} = \text{Mod}_G, \mathcal{B} = \text{Mod}_{G/H}, \mathcal{C} = \text{Ab}$.
- $F(A) = A^H, G(B) = B^{G/H}$.

Then $R^p G = H^p(G/H, -)$, $R^q F = H^q(H, -)$, and $R^{p+q}(G \circ F) = H^{p+q}(G, -)$.

So we have a spectral sequence

$$0 \longrightarrow E_2^{1,0} \longrightarrow H^1 \longrightarrow E_2^{0,1} \xrightarrow{d_2} E_2^{2,0} \longrightarrow H^2$$

which can be rewritten as

$$0 \longrightarrow H^1(G/H, A^H) \xrightarrow{\inf_{G/H}} H^1(G, A) \xrightarrow{\text{res}} H^1(H, A)^{G/H} \xrightarrow{d_2} H^2(G/H, A^H) \longrightarrow H^2(G, A)$$

5.3 Application

Let G be a finite group and A be a G -module.

Theorem 5.5: If $H \leq G$ is a subgroup of index m , then

$$\text{cor}_G^H \circ \text{res}_G^H = m \text{id}_{H^n(G, A)}$$

Proof: For $n = 0$, it is clear.

By dimension shifting, we are done. ■

Corollary 5.6: Take $H = 1$, then we know that $H^n(G, A)$ is of $|G|$ -torsion for $n > 0$.

Corollary 5.7: If A is of finite type, then $H^n(G, A)$ is finite for $n > 0$.

Proof: By the previous corollary, $H^n(G, A)$ is of $|G|$ -torsion.

Also, $H^n(G, A)$ is finitely generated abelian group since A is finitely generated by the cochain complex construction.

So it is finite. ■

Corollary 5.8: Suppose that A is uniquely divisible i.e. A is a $\mathbb{Q}$ -vector space. Then $H^n(G, A) = 0$ for $n > 0$.

Proof: Multiplication by $|G|$ is an automorphism of A since A is uniquely divisible.

But it is zero map on $H^n(G, A)$ for $n > 0$. So $H^n(G, A) = 0$ for $n > 0$. ■

Example 5.9: Consider the exact sequence of G -modules

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Q} \longrightarrow \mathbb{Q}/\mathbb{Z} \longrightarrow 0$$

with the trivial G -action. Then by the long exact sequence of cohomology, we have

$$H^1(G, \mathbb{Q}/\mathbb{Z}) \cong H^2(G, \mathbb{Z})$$

Now suppose G is a Galois group of a finite Galois extension L/K .

Since $\mathbb{Q}/\mathbb{Z}$ is of trivial G -action, we have

$$H^1(G, \mathbb{Q}/\mathbb{Z}) \cong \text{Hom}(G, \mathbb{Q}/\mathbb{Z}) = \text{Hom}(G^{\text{ab}}, \mathbb{Q}/\mathbb{Z})$$

As any finite subgroup of $\mathbb{Q}/\mathbb{Z}$ is cyclic, we have

$$G/\ker f \cong \text{im } f \leq \mathbb{Q}/\mathbb{Z}$$

and $\text{im } f \cong \mathbb{Z}/n\mathbb{Z}$ for some n .

So to give a f is equivalent to giving a cyclic quotient of G and a generator of the cyclic group $\text{Gal}(L^{\ker f}/K)$.

Example 5.10: Same as above. We have $H^1(G, \mathbb{Z}) = 0$.

6 Lecture 2: Tate cohomology

Definition Let G be a finite group.

1. The element norm $\alpha_0 \in \mathbb{Z}[G]$ is defined as

$$\alpha_0 = \sum_{g \in G} g$$

2. Let A be a G -module. The norm map $N : A \rightarrow A$ is defined as

$$N(a) := \alpha_0 a = \sum_{g \in G} ga$$

3. The augmentation (two-sided) ideal I_G is defined as

$$I_G = \ker(\mathbb{Z}[G] \rightarrow \mathbb{Z}) = \text{span}_{\mathbb{Z}}(g - 1 : g \in G)$$

4. $A_G := A/I_G A = \mathbb{Z}[G]/I_G \otimes_{\mathbb{Z}[G]} A$, which is the maximal quotient of A on which G acts trivially.

As the functor $A \mapsto A_G$ is given by a tensor product, it is right-exact.

So we can define its left derived functor, denote by $H_n(G, A)$.

Example 6.1: $H_1(G, \mathbb{Z}) = G^{\text{ab}}$.

Proof: Consider the exact sequence of G -modules

$$0 \longrightarrow I_G \longrightarrow \mathbb{Z}[G] \longrightarrow \mathbb{Z} \longrightarrow 0$$

We have the long exact sequence of homology

$$0 \longrightarrow H_1(G, \mathbb{Z}) \longrightarrow I_G/I_G^2 \longrightarrow \mathbb{Z}[G]_G = \mathbb{Z} \longrightarrow \mathbb{Z}_G = \mathbb{Z} \longrightarrow 0$$

So $H_1(G, \mathbb{Z}) \cong I_G/I_G^2$.

Consider the map $G \rightarrow I_G/I_G^2$ given by $g \rightarrow [g - 1]$.

Since $I_G = \text{span}_{\mathbb{Z}}(g - 1 : g \in G)$, the map is surjective.

Also, since I_G/I_G^2 is abelian, the map factors through G^{ab} .

Now consider the map $I_G \rightarrow G^{\text{ab}}$ given by $g - 1 \rightarrow g$.

It is surjective and factors through I_G/I_G^2 .